



Assignment 7.1: Security Hardening Report with Updated System Design and Drawings

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Executive Summary

Four Point Cyber Consulting completed this Security Hardening Report for Design World following the approved Security Hardening Plan and Project Change Requests. The report documents the hardening actions completed across key systems supporting Design World's remote access, identity services, internal file storage, workstation operations, and web services. These actions were based on vulnerabilities confirmed during prior assessment and penetration testing, including weak authentication, missing patches, insecure file sharing, vulnerable packages, and unencrypted remote access.

Hardening was completed on the Domain Controller, File Server, Ubuntu Workstation, and CentOS Web Server. The work included applying security updates, disabling legacy authentication, enforcing SMB signing, restricting NFS exposure, removing Telnet, reviewing service permissions, and confirming secure SSH-based administration. The Transit VLAN and OpenVPN Server hardening remains open because administrative access to DW-OpenVPN-Server-OL-1 was not available during implementation. Once access is provided, Design World should complete the remaining Telnet decommissioning, SSH restriction, firewall review, and logging validation steps.

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1. Scope

This Security Hardening Report documents the approved security improvements implemented for Design World’s representative enterprise cloud environment. Design World is an international structural design company that relies on secure remote access, centralized identity services, protected file storage, workstation access, web services, and internal network routing to support business operations. The scope is limited to the controlled Design World mirror environment, including the Public Subnet, Business Subnet, Website/DMZ Subnet, and Transit Services VLAN. The systems included are the OpenVPN Server, Windows Server 2022 Domain Controller, Windows Server 2022 File Server, Ubuntu 22.04 LTS Workstation, and CentOS Stream 9 Web Server. Hardening activities include approved patching, configuration updates, service hardening, authentication improvements, firewall review, access control updates, logging improvements, and validation testing.

2. Objectives

The objective of this report is to summarize the implementation of approved security hardening measures that reduce the risks confirmed during Design World’s vulnerability assessment and penetration test. These risks included missing security updates, weak authentication settings, insecure service permissions, SMB weaknesses, world-readable NFS shares, outdated OpenSSL packages, exposed Telnet services, and insufficient restrictions on administrative access. The hardening effort is intended to reduce the likelihood of unauthorized access, credential exposure, privilege escalation, lateral movement, insecure file access, and service compromise while maintaining system availability for authorized users.

3. Background

Design World depends on its cloud-based infrastructure to support remote employees, internal business operations, customer-facing services, and the protection of sensitive design information. Prior vulnerability assessment and penetration testing activities identified weaknesses across key systems, including the Domain Controller, File Server, Ubuntu Workstation, CentOS Web Server, OpenVPN Server, and Transit Services VLAN. To address these issues, Four Point Cyber Consulting prepared a Security

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Hardening Plan and five Project Change Requests for approval through the change management process. This report documents the move from planning to implementation by summarizing the approved hardening actions taken to strengthen Design World's environment and prepare it for post-hardening validation.

4. Completed Hardening Measures

Security hardening measures were completed across the approved Design World systems where access was available. The completed work focused on reducing the risks confirmed during vulnerability assessment and penetration testing, including weak authentication, insecure service permissions, missing patches, SMB exposure, world-readable file sharing, vulnerable Linux packages, and unencrypted remote access. The Domain Controller, File Server, Ubuntu Workstation, and CentOS Web Server received configuration updates, patching, service hardening, and validation checks. The Transit VLAN and OpenVPN Server hardening could not be completed because direct administrative access to DW-OpenVPN-Server-OL-1 was not available during the implementation period.

4.1 Domain Controller Hardening

Security hardening measures were completed on the Windows Server 2022 Domain Controller, DW-DC-OL-1, to reduce risks related to authentication, patching, service permissions, and cryptographic validation. Legacy LM and NTLMv1 authentication were disabled, and stronger NTLMv2 and Kerberos authentication were enforced where applicable. This change reduced exposure to password hash attacks and credential-based attacks associated with older authentication protocols.

Additional hardening was completed by enabling WinVerifyTrust signature validation through the EnableCertPaddingCheck registry setting. This improves protection against improperly modified signed files being trusted by the system. Windows service permissions were also reviewed to identify insecure discretionary access control lists that could support local privilege escalation. Missing Windows security updates were applied through Windows Update, including operating system updates, .NET Framework updates, and Microsoft Defender security intelligence updates. These changes

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strengthened the Domain Controller and reduced risk to Design World’s centralized identity services.

4.2 File Server Hardening

Security hardening measures were completed on the Windows Server 2022 File Server, DW-FS-OL-1, to protect Design World’s internal file storage and reduce unauthorized access risk. SMB signing was enforced to help prevent SMB relay and man-in-the-middle attacks. Legacy LM and NTLMv1 authentication were disabled, and stronger authentication methods were applied to improve access control.

File share permissions were reviewed, and world-readable NFS exposure was restricted to authorized users and approved hosts. Missing critical Windows Server updates were also applied to address known operating system vulnerabilities. These changes reduced the File Server’s attack surface, improved the confidentiality and integrity of stored business data, and strengthened the system supporting Design World’s internal file access.

4.3 Ubuntu Workstation Hardening

Security hardening measures were completed on the Ubuntu 22.04 LTS Workstation, DW-U22-OL-1, to reduce exposure to vulnerable packages and insecure remote management. Package hold settings were reviewed and removed where possible so that security updates could be applied. System packages were updated, including OpenSSL and related dependencies, to remediate known software vulnerabilities.

Telnet services were disabled and removed to prevent unencrypted remote access. SSH remained the approved method for remote administration, and the SSH configuration was reviewed for secure communication. During validation, the OpenSSL version appeared to remain at 3.0.2, which may limit full vulnerability remediation depending on package availability and repository support. Design World should continue tracking this condition and either allow approved package version updates or document compensating controls if the version cannot be upgraded further.

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4.4 CentOS Web Server Hardening

Security hardening measures were completed on the CentOS Stream 9 Web Server, DW-CS9W-OL-1, to strengthen the Website Subnet and reduce web server exposure. Available security updates were applied to remediate vulnerable packages and dependencies. The Apache web server configuration was reviewed to identify unnecessary exposure and confirm that only approved web services and ports were available.

Secure remote administration was also reviewed by confirming that SSH remained the approved administrative protocol. Unnecessary or unauthorized management services were checked to reduce the chance of exposed administrative access. These combined actions reduced the CentOS web server's attack surface, improved patch management, and strengthened the system supporting Design World's public-facing web services.

4.5 Transit VLAN and OpenVPN Server Hardening Status

The Transit VLAN and OpenVPN Server hardening for DW-OpenVPN-Server-OL-1 was not completed because administrative access to the system was not available during implementation. The planned hardening activity was to disable unencrypted Telnet server daemons, block Telnet access on port 23, require encrypted remote administration through SSH, restrict SSH access to approved OpenVPN client sources, review firewall rules, and improve authentication logging.

Because access was unavailable, Four Point Cyber Consulting could not validate the current Telnet service status, modify firewall rules, restrict SSH access, or confirm logging changes on DW-OpenVPN-Server-OL-1. This item should remain open as a deferred hardening task. Once access is restored or provided by Design World, the team should complete the planned management protocol decommissioning and perform post-change validation to confirm that Telnet is disabled and that approved SSH access remains available.

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5. Approved Project Change Requests (PCRs) Implementation

Following approval of the Project Change Requests, Four Point Cyber Consulting implemented the approved hardening measures on Design World systems where administrative access was available. The PCRs were used to document and control changes affecting identity services, file storage, workstation security, web server configuration, and the OpenVPN and Transit VLAN administrative access path. PCR-001 through PCR-004 were completed, while PCR-005 remains deferred because administrative access to DW-OpenVPN-Server-OL-1 was not available during the implementation period.

5.1 PCR-001: Windows Server 2022 Domain Controller Hardening

Summary of Approved Changes

PCR-001 approved hardening for DW-DC-OL-1 by addressing missing security updates, weak authentication settings, WinVerifyTrust signature validation, and insecure service permissions.

Configuration and Security Updates

The Domain Controller was updated to disable legacy LM and NTLMv1 authentication, enforce stronger NTLMv2 and Kerberos authentication where applicable, enable WinVerifyTrust mitigation, review insecure service permissions, and apply missing Windows Server security updates.

Impact

These changes strengthened Design World's centralized identity services by reducing the risk of credential attacks, unauthorized access, improper trust of modified signed files, and privilege escalation.

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5.2 PCR-002: Windows Server 2022 File Server Hardening

Summary of Approved Changes

PCR-002 approved hardening for DW-FS-OL-1 to improve the protection of Design World's internal file storage and reduce exposure caused by weak SMB, legacy authentication, missing patches, and insecure file sharing.

File Server and Access Control Updates

Missing security updates were applied, SMB signing was enforced, legacy LM and NTLMv1 authentication were disabled, and world-readable file sharing was restricted to authorized users and approved hosts.

Impact

These changes improved the confidentiality and integrity of Design World's stored data by reducing the risk of unauthorized file access, SMB relay attacks, man-in-the-middle activity, and exposure of sensitive business and design files.

5.3 PCR-003: Ubuntu 22.04 Workstation Hardening

Summary of Approved Changes

PCR-003 approved hardening for DW-U22-OL-1 by addressing vulnerable packages, package hold settings, Telnet exposure, and SSH-based administration.

System and Remote Administration Updates

Package restrictions were reviewed, system packages were updated, Telnet was disabled and removed, and SSH remained the approved method for remote administration. OpenSSL should continue to be monitored because the available version may remain limited by package repository support.

Impact

These changes reduced exposure to known software vulnerabilities and removed unencrypted remote access from the Ubuntu workstation while preserving secure administrative access through SSH.

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5.4 PCR-004: CentOS Stream 9 Web Server Hardening

Summary of Approved Changes

PCR-004 approved hardening for DW-CS9W-OL-1 by addressing system updates, vulnerable packages, Apache configuration, and secure remote administration.

Web Server and Security Updates

CentOS security updates were applied, vulnerable packages and dependencies were updated, the Apache web server configuration was reviewed, approved web services and ports were confirmed, and SSH was validated as the secure administrative method.

Impact

These changes reduced the web server's attack surface, improved patch management, and strengthened the Website Subnet while maintaining required web service availability.

5.5 PCR-005: OpenVPN and Transit VLAN Hardening

Summary of Approved Changes

PCR-005 approved hardening for the OpenVPN Server and Transit Services VLAN by disabling Telnet, blocking port 23, requiring SSH for remote administration, restricting SSH access, reviewing firewall rules, and improving authentication logging.

Implementation Status

PCR-005 was not fully implemented because administrative access to DW-OpenVPN-Server-OL-1 was unavailable during the hardening period. Because access was not available, Four Point Cyber Consulting could not validate the Telnet service status, modify firewall rules, restrict SSH access, or confirm VPN and SSH logging changes.

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Impact / Remaining Risk

The impact of PCR-005 remains pending until access is provided. The remaining risk includes possible Telnet exposure, incomplete SSH restriction, unvalidated firewall rules, and incomplete logging verification. Once access is restored, Design World should complete this PCR and validate that Telnet is disabled, port 23 is blocked, SSH is restricted to approved sources, and VPN and SSH authentication activity is logged.

6. Updated System Design and Drawings

The updated Design World system design keeps the same segmented architecture while reflecting the security hardening measures completed during implementation. The Business Subnet continues to support the Domain Controller, File Server, and Ubuntu Workstation. The Website/DMZ Subnet continues to support the CentOS Stream 9 Web Server. The Public Subnet and Transit Services VLAN remain responsible for VPN access, routing, and administrative traffic between network zones.

The post-hardening design reflects stronger identity, file sharing, workstation, and web server controls. The Domain Controller now uses stronger authentication settings and improved signature validation. The File Server now enforces SMB signing and restricts insecure file sharing. The Ubuntu Workstation and CentOS Web Server now rely on SSH for secure administration, with Telnet removed where access was available. The OpenVPN Server and Transit VLAN should be marked as pending because administrative access to DW-OpenVPN-Server-OL-1 was not available during implementation.

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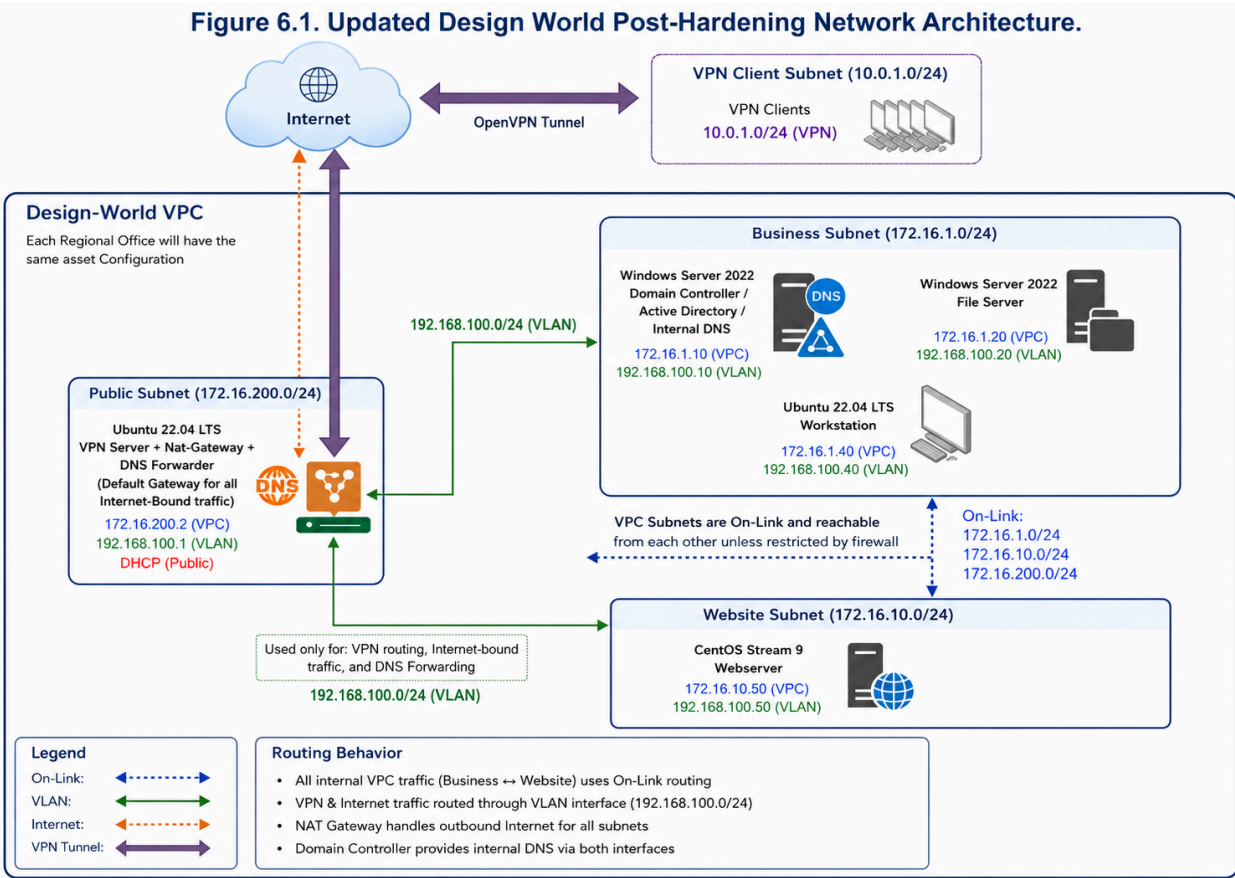


Figure 6.1. Updated Design World Post-Hardening Network Diagram

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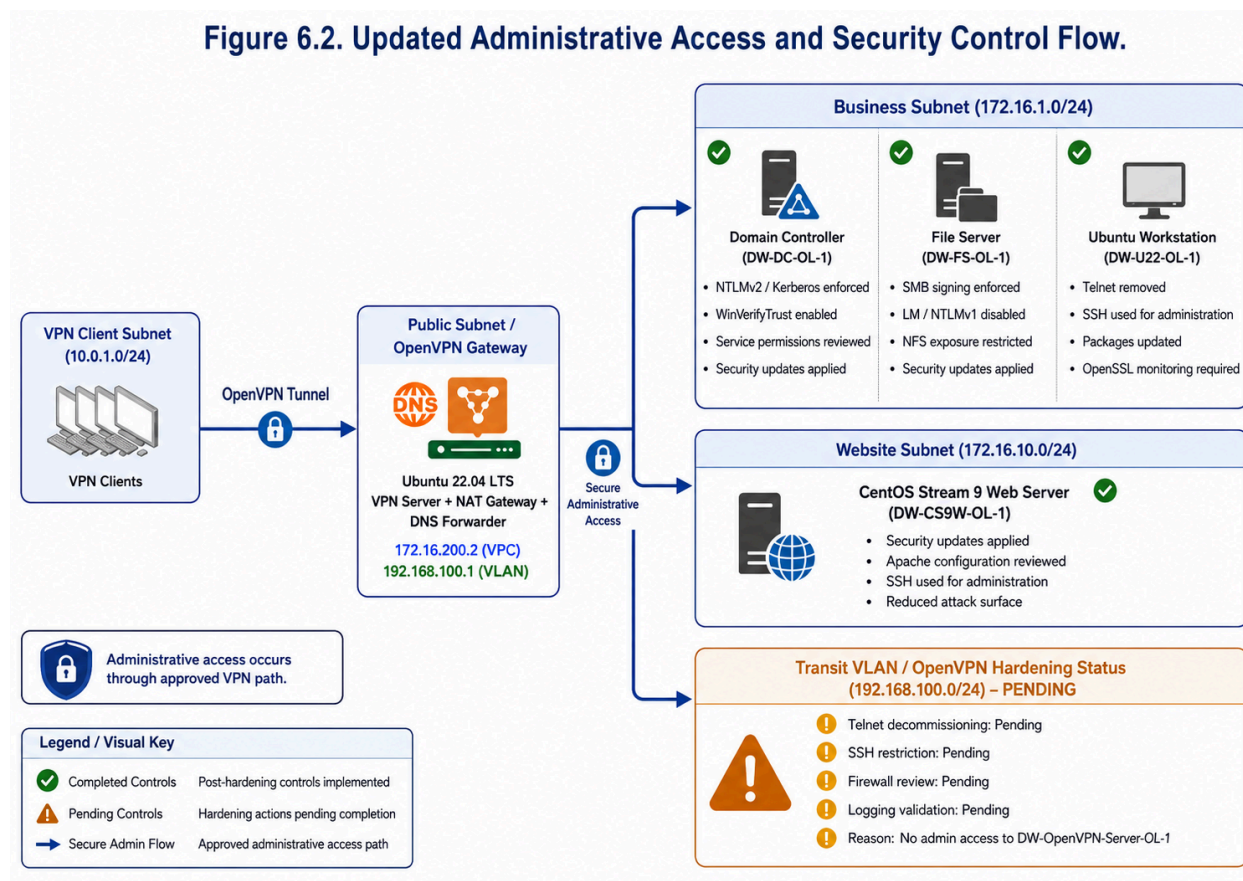


Figure 6.2. Updated Administrative Access and Security Control Flow

7. Conclusion and Next Steps

The approved Security Hardening Plan and Project Change Requests provide Design World with a structured approach for reducing confirmed security risks across its enterprise cloud environment. The implementation focused on the systems most important to business operations, including identity services, internal file storage, workstation administration, web services, VPN access, and Transit VLAN routing.

The next step is to complete post-implementation validation. Four Point Cyber Consulting will verify that security updates were applied, insecure services were disabled, authentication settings were strengthened, SMB signing was enforced, NFS

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access was restricted, firewall rules were reviewed, and administrative logging was improved. Validation should include configuration review, patch verification, service testing, access testing, vulnerability rescanning, and screenshot evidence.

After validation is complete, the team should move into the post-hardening assessment phase. This phase should include the Post-Hardening Vulnerability Assessment Report, Post-Hardening Penetration Test Report, Design World Information System Security Plan, Final Project Report, and Final Presentation. These deliverables will help confirm whether the implemented hardening measures improved Design World's security posture and reduced the likelihood that previously confirmed vulnerabilities can be exploited.

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8. Approvals

Role	Printed Name	Signature	Date
Project Lead	Christian Calipusan	<i>Christian Calipusan</i>	06/22/26
Security Analyst	Shivam Patel	<i>Shivam Patel</i>	06/22/26
Security Engineer	Joshua Remington	<i>Joshua Remington</i>	06/22/26
Network Engineer	Mikailou Nikiema	<i>Mikailou Nikiema</i>	06/22/26
DW CEO			